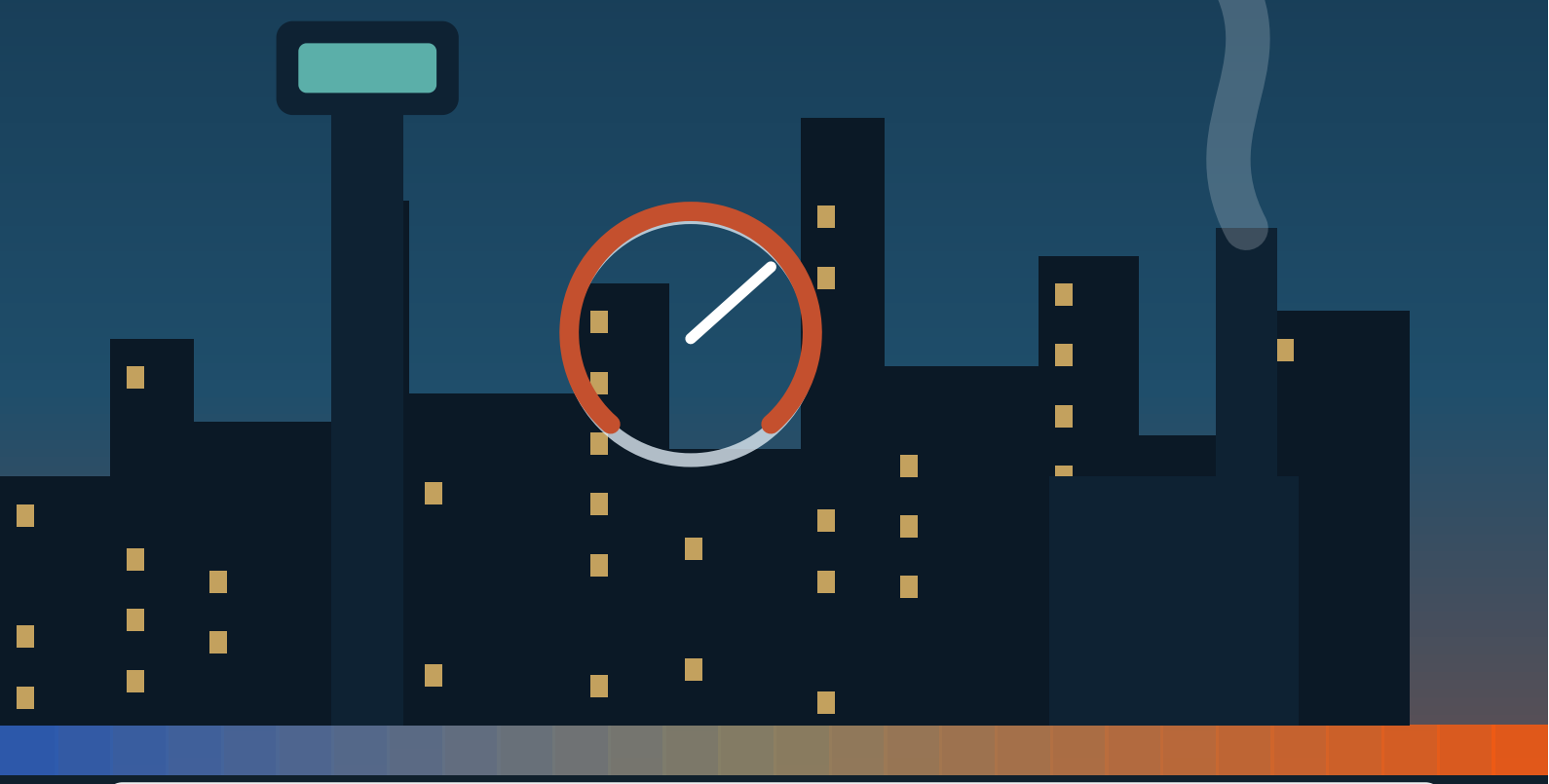
GO 1 OF 3 · MODELLING CLIMATE AND ENERGY FUTURES

See the whole picture

Every 1.5 °C pathway in the IPCC's 2022 report came out of a computer model. This guide shows you, in pictures and with your own hands, how those models think.



INSIDE

- Why a model of the year 2100 is a **what-if machine**, not a crystal ball
- The **bathtub** that explains carbon budgets in one picture
- Why **$100 of damage in 2125** is worth about $5 today
- A guided tour of **GCAM**, the model behind one of the new CMIP7 scenarios
- Fifteen **predict-then-check** activities with free online simulators

A beginner's study guide · no maths beyond percentages · October 2026

The map

Integrated assessment models (IAMs) are the computer models behind the climate pathways you see in IPCC reports, national net-zero plans and bank stress tests. This guide is the first of three passes over the subject. Here you see the whole machine from the outside; Go 2 opens each box; Go 3 uses one model, GCAM, for research.

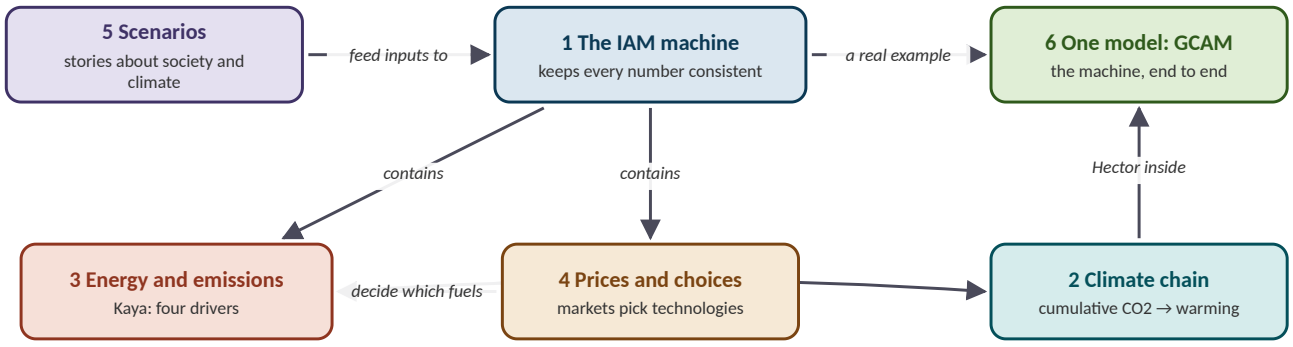


Figure 0 — the six ideas of Go 1 and how they connect. Notice that scenarios sit outside the machine: they are what you feed in.

HOW TO USE THIS GUIDE

1. Read one concept per sitting (about 40 minutes).
2. Do every **Predict, then check** box: write your guess *before* you open the simulator.
3. Walk the memory route at the end of each concept.
4. Finish with the quiz; it tells you which page to reread.

WHAT YOU WILL BE ABLE TO DO

- Say in two sentences what an IAM is and what it is not.
- Explain why warming follows *cumulative* CO2.
- Break emissions into four drivers.
- Explain what a carbon price and a discount rate do.
- Decode a label such as SSP2-4.5 or C3.
- Read any IAM result as Policy − Reference.

Contents: 1 What an IAM is | 2 The climate chain | 3 Energy and emissions | 4 Prices and choices | 5 Scenarios | 6 One model, end to end

Where the explorers live. Course explorers are on the course website, in the Explore tab of the week named in each box. External simulators are free; a few IIASA explorers show a log-in link, so try them once before class. All web addresses were checked on 23 October 2026.

CONCEPT 1 OF 6

What an integrated assessment model is

A consistent what-if machine, not a crystal ball

HOOK A weather forecast is unreliable about two weeks ahead, yet IPCC reports are full of charts that run to 2100. For its 2022 mitigation report the IPCC collected about 3,000 such pathways and sorted 1,202 of them by how much warming they lead to. None of them is a forecast. So what are they?



An **integrated assessment model** joins the pieces of the climate problem into one calculation: the economy, energy, land and water, the emissions they produce, and the climate that responds. **Integrated** means a change in one part shows up in the others. **Assessment** means its purpose is to inform decisions.

Think of a **recipe calculator**. You type in ingredients you *choose*: how many people, how rich, how cheap solar is, which policies exist. The calculator returns a cake that adds up: energy used equals energy made, emissions match the fuel burned, warming matches the emissions. Where the analogy breaks: a real cake can be tasted; a 2100 world cannot, so you judge a model by its consistency and its assumptions, not by the result.

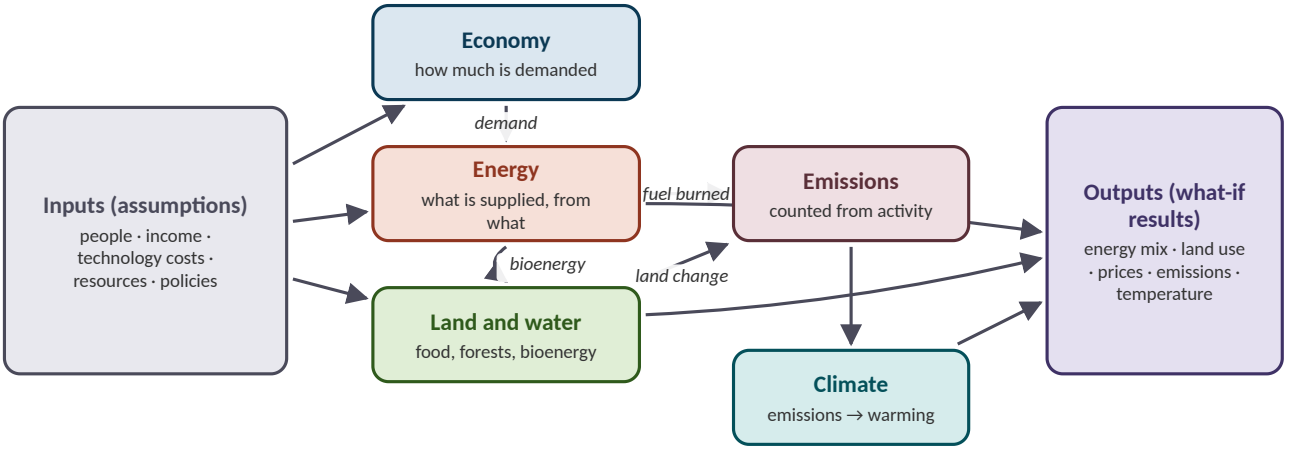


Figure 1.1 — the IAM machine. What to notice: *policies are an input*, chosen by the user; temperature is an output. Arrows show which part sends information to which.

Every box in Figure 1.1 must agree with every other box. That single demand, **consistency**, is what an IAM adds that a spreadsheet of separate guesses does not.

$$\mathrm{CO_2} = \text{energy used} \times \mathrm{CO_2} \text{ per unit of energy} \tag{1}$$

SIMPLE CALCULATION

Why consistency matters (illustrative numbers from week 1)

1. The world uses 600 units of energy a year; each unit releases 0.06 Gt of CO_2.
2. Before: 600 × 0.06 = 36 Gt of CO_2.
3. A policy cleans up the energy mix so each unit releases 0.03 Gt.
4. After: 600 × 0.03 = 18 Gt. Change: 36 − 18 = 18 Gt.
5. As a share: 18 ÷ 36 = 0.5, a 50% cut. A consistent model cannot claim the cut *and* keep the old emissions.

Answer: emissions fall by 18 Gt, a 50% cut

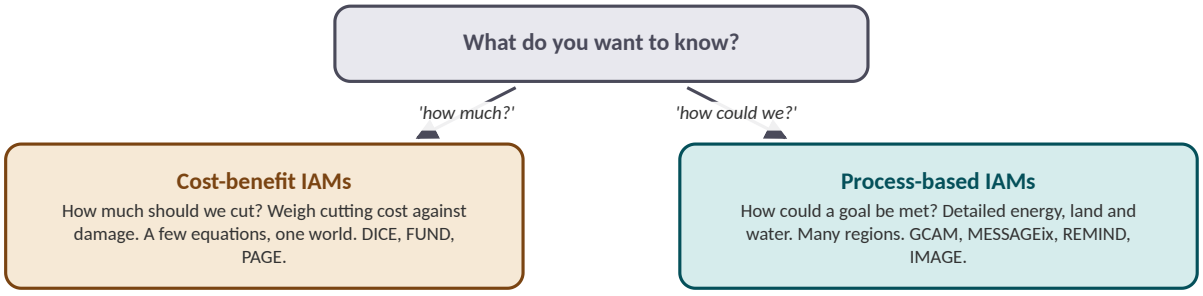


Figure 1.2 — two families of IAMs. What to notice: they answer different questions. GCAM, the model of Go 2 and Go 3, is process-based.

TRAP "The model predicts 2.7 °C by 2100." No: it **projects** what follows *if* its assumptions hold. Change the assumptions and the number changes. And an IAM does not decide policy: policies are inputs the user chooses.

PICTURE THIS · STOP 1 OF THE ROUTE At the **city gate** stands a giant brass **recipe machine.** Into its hopper you throw little wooden people, gold coins, a solar panel and a law book (the inputs). The machine shakes, every gauge on its side swings into line, and out slides a cake with a **thermometer candle** (the outputs). Two gatekeepers argue beside it: one holds scales (cost-benefit), the other a toolbox (process-based).

PREDICT, THEN CHECK

In the course **Warming explorer** keep the peak in 2025. If CO_2 reaches zero in **2070** instead of **2050**, will 2100 warming be higher or lower, and by about how much?

My prediction	What the explorer showed

Explorer: Warming explorer (course site → Explore tab of week 1). **Check:** higher: about 1.63 °C instead of 1.44 °C in this toy (+0.19 °C), because 20 more years of emissions add to the running total

PREDICT, THEN CHECK

Open **En-ROADS** (https://en-roads.climateinteractive.org). Before touching anything, write down which *one* slider you think moves 2100 warming the most. Then try it, and then try a second slider.

My prediction	What the explorer showed

Explorer: En-ROADS, Climate Interactive and MIT Sloan, free, no log-in. **Check:** write down the temperature shown after each move; compare with your guess. Notice that the model rebalances energy, prices and emissions every time: that is consistency at work

TRY IT NOW In one minute, sort these into **inputs** or **outputs** of an IAM: (a) population in 2050, (b) coal used in 2050, (c) a carbon price, (d) warming in 2100, (e) the cost of solar panels, (f) the price of electricity.

LOOK IT UP

- Carbon Brief, Q&A: How integrated assessment models are used to study climate change (read the first half).
- IAMC, What are IAMs? — the community's own definition and the two families.
- Resources for the Future podcast, Understanding climate models, with Massimo Tavoni (audio with transcript).

RECAP SO FAR

1. **What an IAM is:** a consistent what-if machine; inputs are assumptions; outputs are projections, not predictions.

CONCEPT 2 OF 6

The climate chain

Warming follows the running total of CO2

HOOK If the world stopped emitting CO_2 tomorrow, the warming caused by CO_2 would roughly **stop rising**; it would not keep climbing for centuries. The flip side: the warming already caused does not go away either. Both facts come from one idea, the running total.



Every IAM ends with the same chain (Figure 2.1). Burning fuel and clearing land puts CO_2 into the air. More CO_2 in the air (the **concentration**, in parts per million, ppm) traps extra heat (the **forcing**, in watts per square metre). Extra heat warms the planet. CO_2 stays in the air for centuries, so what matters is not this year's emissions but the **total ever emitted**.



Figure 2.1 — the climate chain. What to notice: an IAM computes the first box itself and hands it to a simple climate model for the next three.

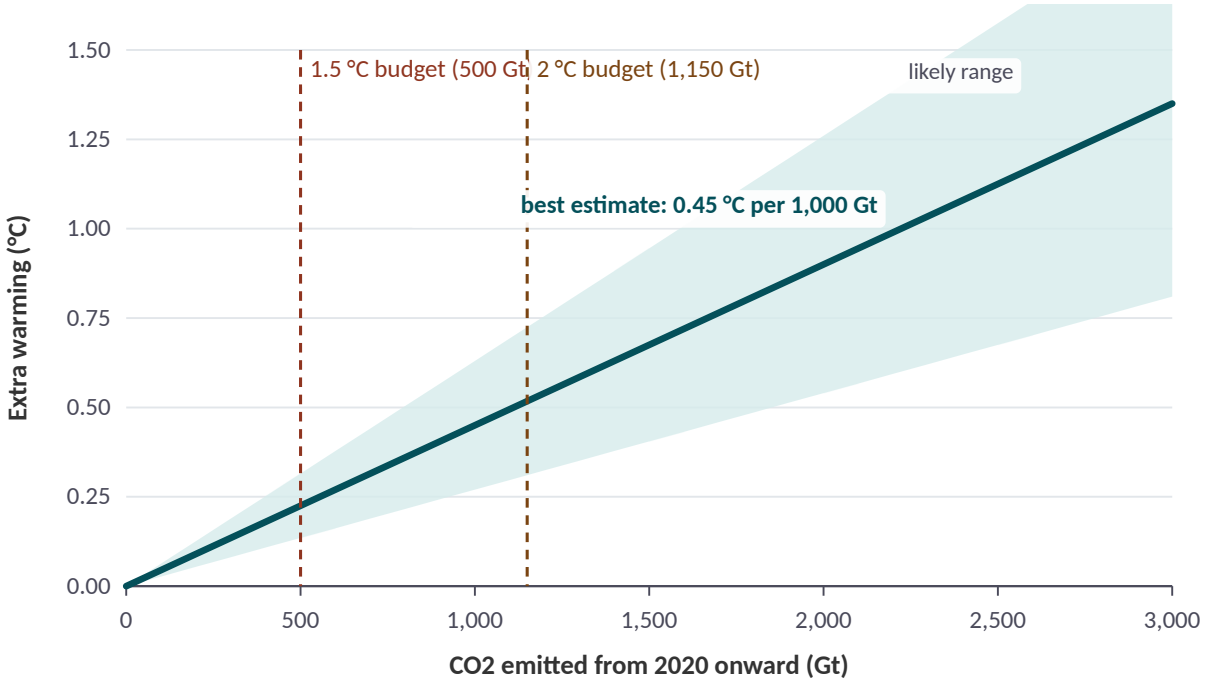


Figure 2.2 — warming is a straight line in cumulative CO_2. What to notice: the line is straight, so every 1,000 Gt adds about the same warming, wherever you are on it. The shaded band is the IPCC's likely range (0.27 to 0.63 °C per 1,000 Gt).

$$\text{extra warming} \approx 0.45\ ^\circ\text{C} \times (\text{CO}_2\ \text{emitted} \div 1{,}000\ \text{Gt}) \tag{2}$$

Picture a **bathtub** (Figure 2.3). The tap is yearly emissions, about 40 Gt of CO_2. The drain is almost blocked, because nature removes CO_2 only slowly. The rim is the **carbon budget**: how much more can go in before a temperature limit is crossed. Where the analogy breaks: a real bath can be bailed out quickly; removing CO_2 from the air is slow and costly (concept 6).

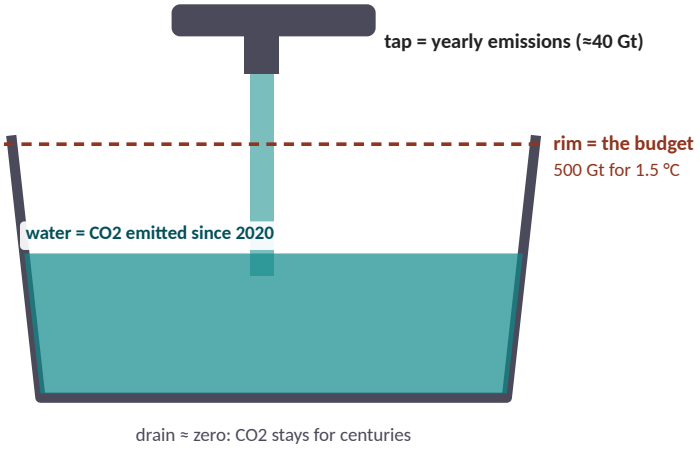


Figure 2.3 — the carbon budget bathtub. What to notice: turning the tap down slows the rise but does not lower the water. Only the total matters for the rim.

SIMPLE CALCULATION

How long does the 1.5 °C budget last?

1. Remaining budget for a 50% chance of 1.5 °C, counted from 2020: 500 Gt (IPCC AR6).
2. If emissions stay flat at 40 Gt a year: 500 ÷ 40 = 12.5 years, then a sudden stop.
3. If emissions fall in a straight line to zero over T years, the total is a triangle: ½ × 40 × T.
4. Set ½ × 40 × T = 500, so 20 × T = 500 and T = 25 years.
5. For 2 °C (1,150 Gt): flat lasts 1,150 ÷ 40 = 28.8 years.

Answer: 12.5 years flat; 25 years with a steady straight-line fall to zero

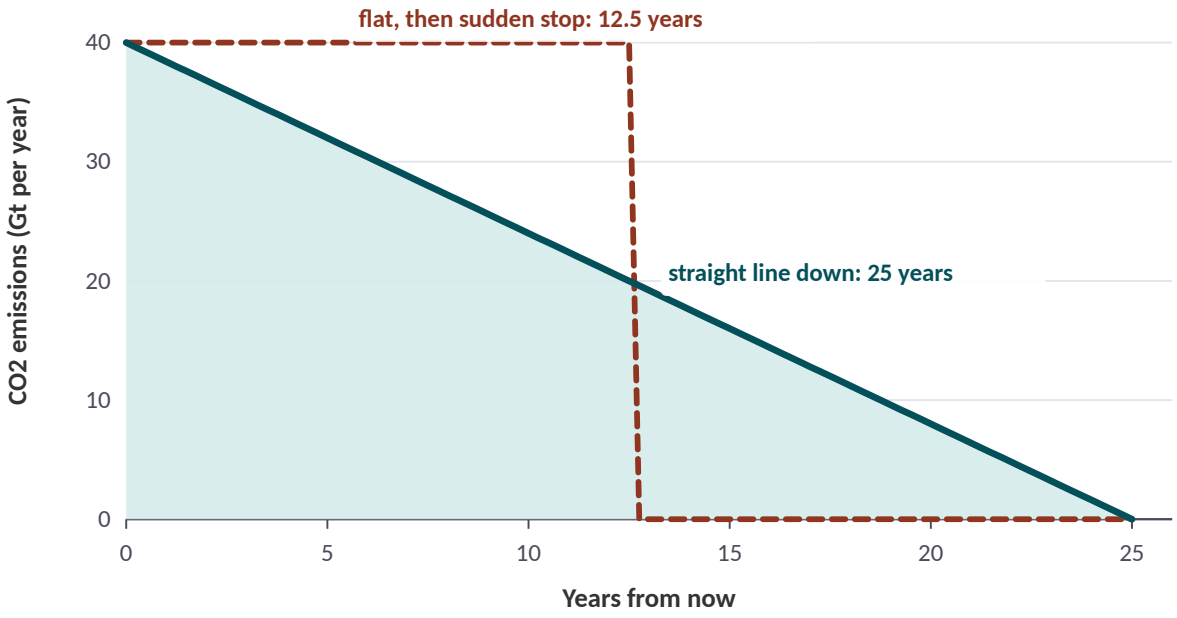


Figure 2.4 — same budget, two shapes. What to notice: the two areas under the lines are equal (500 Gt). A gradual fall buys twice the time of a sudden stop.

TRAP "Emissions are falling, so warming is falling." No: while emissions are above zero the bathtub still fills, so CO_2 and temperature still rise, just more slowly. And **net zero** is not zero: it means what is still emitted equals what is removed.

PICTURE THIS · STOP 2 OF THE ROUTE In the square stands a **bathtub fountain**. A tap pours exactly **40 buckets** a year; the plughole is stuffed with a cork. A red line painted at the rim reads **500**. Beside it, a thermometer is glued to the water level: every time the water rises one notch (1,000 buckets), the mercury climbs **0.45** degrees.

PREDICT, THEN CHECK

Course **Carbon budget bathtub**: set the goal to 1.5 °C and emissions to **30 Gt** a year. How many years does the budget last if emissions stay flat? And if they fall in a straight line?

My prediction	What the explorer showed

Explorer: Carbon budget bathtub (course site → Explore tab of week 2). **Check:** 16.7 years flat, 33.3 years in a straight line (500 ÷ 30 and twice that)

PREDICT, THEN CHECK

Open Our World in Data, CO_2 and Greenhouse Gas Emissions. Guess which country has emitted the most CO_2 *in total since 1750*, and whether it is the same country that emits most today.

My prediction	What the explorer showed

Explorer: Our World in Data CO2 data explorer, free, CC BY. **Check:** switch the explorer between annual and cumulative emissions. The ranking changes: today's largest emitter is not the largest in the running total, which is the quantity the climate responds to

TRY IT NOW Use Equation 2: if the world emits 800 Gt more, how much extra warming is that? And how many years of today's emissions is 800 Gt?

LOOK IT UP

- IPCC AR6 Synthesis Report, Figure SPM.1 — warming stripes for people born in 1950, 1980 and 2020.
- Carbon Brief, Q&A: How do climate models work? — why an IAM's climate part is a simple model.
- MAGICC live, https://live.magicc.org — a simple climate model in the browser (instructor demo; access not confirmed).

RECAP SO FAR

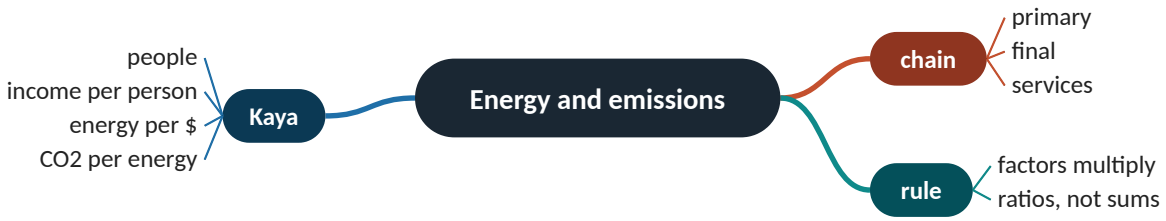
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CONCEPT 3 OF 6

Energy and emissions

CO2 = people × income × energy per $ × CO2 per energy

HOOK Of every 100 units of energy in the coal fed to a typical power plant, only about 36 reach your home as electricity. So saving one unit at home saves about 2.8 units of coal (illustrative efficiencies: 40% plant, 10% grid losses).



People do not want energy; they want **services**: light, warmth, travel, steel. Energy travels a chain to get there (Figure 3.1). **Primary energy** is what comes out of the ground or the sky. **Conversion** (power plants, refineries) turns it into **final energy**: electricity, petrol, heat. Every step loses some.

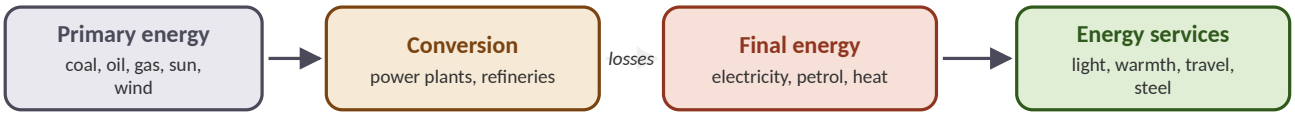


Figure 3.1 — from primary energy to services. What to notice: process-based IAMs choose technologies at each step, so they can compare cleaner supply with lower demand.

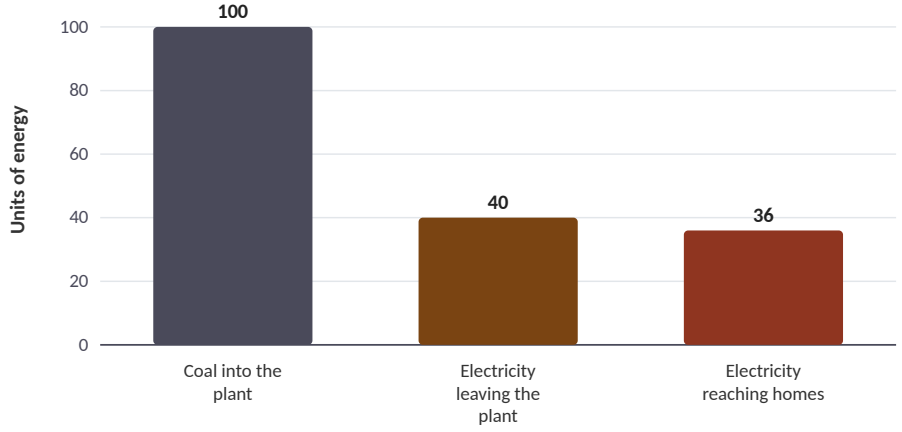


Figure 3.2 — energy lost along the chain (illustrative). What to notice: most of the loss happens inside the power plant, before the grid.

To see what drives emissions, split them into four factors that multiply. This is the **Kaya identity**, named after the energy economist Yoichi Kaya. It is not a theory: the units cancel, so it is always true. Its value is as a way of sorting the data.

$$\mathrm{CO_2} = \text{people} \times (\text{GDP} \div \text{people}) \times (\text{energy} \div \text{GDP}) \times (\mathrm{CO_2} \div \text{energy}) \tag{3}$$

The first two factors are society: how many people and how rich. The last two are technology: how much energy each dollar needs and how dirty that energy is. IAMs spend most of their detail on the last two, because that is where policy can act.

SIMPLE CALCULATION

Avalon, an invented country, over 30 years

1. People grow 20%: ratio 1.2. Income per person grows 50%: ratio 1.5.
2. Energy per dollar falls 30%: ratio 0.7. CO_2 per unit of energy falls 20%: ratio 0.8.
3. Multiply the ratios: 1.2 × 1.5 = 1.8; 1.8 × 0.7 = 1.26; 1.26 × 0.8 = 1.008.
4. So CO_2 ends 1.008 times its start: up 0.8%, almost flat.

Answer: CO2 rises by under 1%: growth in people and income is almost exactly cancelled by cleaner technology

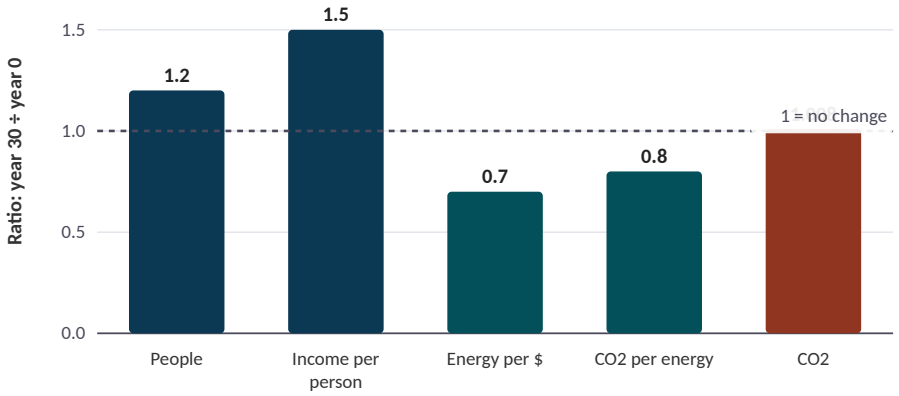


Figure 3.3 — Avalon's four factors. What to notice: two bars above 1 (society) and two below 1 (technology) multiply to a CO_2 bar sitting almost exactly on 1.

TRAP Adding the percentages gives +20 + 50 − 30 − 20 = +20%: wrong. Changes in factors **multiply**, so you multiply the ratios: ×1.008, about +0.8%.

PICTURE THIS · STOP 3 OF THE ROUTE Next stop is a **power station with four huge gears** in a row: a crowd of people turning the first, a pile of coins the second, a dripping leaky pipe the third, and a sooty chimney brush the fourth. The gears are geared together, so they **multiply**. Above the door a sign says **100 in, 36 home**.

PREDICT, THEN CHECK

Course **Kaya calculator**: start from Avalon (+20%, +50%, −30%, −20%). Predict the CO_2 change if energy per dollar falls by 50% instead of 30%.

My prediction	What the explorer showed

Explorer: Kaya calculator (course site → Explore tab of week 3). **Check:** 1.2 × 1.5 × 0.5 × 0.8 = 0.72, so CO2 falls by 28%

PREDICT, THEN CHECK

Open the Kaya Identity Scenario Prognosticator, https://forecast.uchicago.edu/kaya/kaya.html. Predict what happens to 2100 emissions if you set the growth of energy per dollar and CO_2 per energy to zero (no technology progress).

My prediction	What the explorer showed

Explorer: Kaya Identity Scenario Prognosticator, University of Chicago, free, no log-in. **Check:** emissions then grow as fast as people × income, so they climb steeply; the tool also reports how much carbon-free energy would be needed to stabilise CO2

PREDICT, THEN CHECK

Open the Our World in Data chart Change in energy use and GDP per person and pick the United Kingdom. Will the two lines move together or apart since 2000?

My prediction	What the explorer showed

Explorer: Our World in Data Grapher, free, CC BY. **Check:** look for whether income per person rises while energy per person falls; if so, energy per dollar has fallen: the third Kaya factor at work

TRY IT NOW A country's population doubles and its income per person doubles. Energy per dollar and CO_2 per energy both halve. What happens to its CO_2?

LOOK IT UP

- Penn State METEO 469, The Kaya Identity (CC BY-NC-SA).
- Our World in Data, Energy — the same quantities IAMs output.
- Our World in Data, Is the world making progress in decarbonizing energy?

RECAP SO FAR

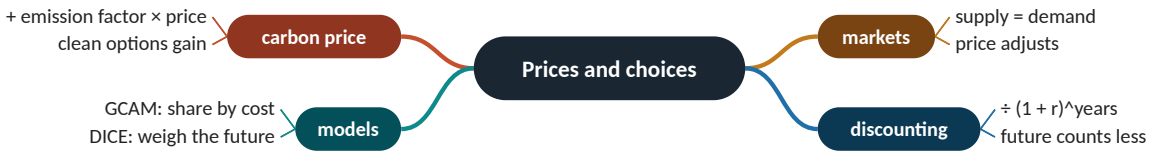
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3. **Energy and emissions:** CO2 = people × income per person × energy per $ × CO2 per energy; the factors multiply.

CONCEPT 4 OF 6

Prices and choices

Prices balance markets, a carbon price changes who wins, discounting weighs the future

HOOK At a 3% discount rate, $100 of climate damage in 100 years is worth about **$5.20** today. At 1.5% it is worth $22.56. One number chosen by the modeller changes the value of the future more than fourfold.



Only three ideas from economics are needed for this course. **One:** a market price keeps moving until what is offered equals what is wanted, like a farmers' market at closing time. That price is the **equilibrium**. GCAM finds such a price for every fuel, crop and region at once.

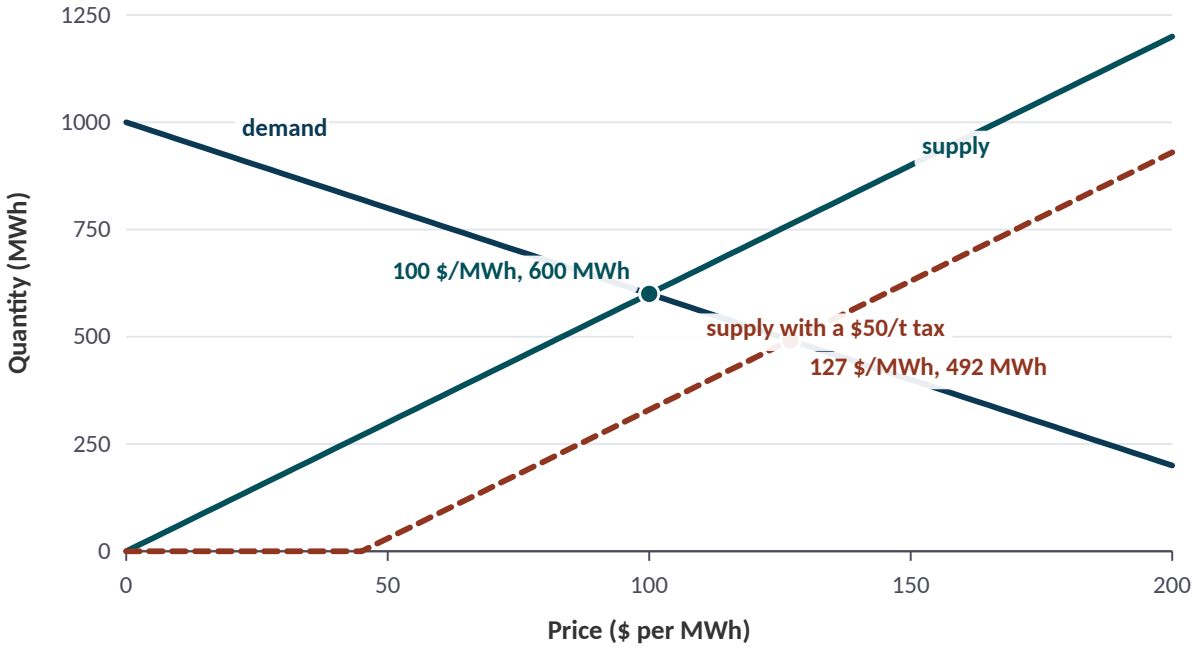


Figure 4.1 — supply and demand (lab 4 toy numbers). What to notice: a tax on coal power shifts supply to the right, so the price rises from 100 to 127 and the quantity sold falls from 600 to 492.

Two: a **carbon price** adds a charge to every option in proportion to its emissions. Solar emits nothing, so its cost does not change; coal is hit hardest.

$$\text{new cost} = \text{start cost} + \text{emission factor} \times \text{carbon price} \tag{4}$$

Technology	Cost, $/MWh	t CO_2 per MWh	With $50 per tonne
Coal	60	0.9	60 + 0.9 × 50 = 105
Gas	70	0.4	70 + 0.4 × 50 = 90
Solar	80	0	80 + 0 × 50 = 80

SIMPLE CALCULATION

When does gas beat coal?

1. Coal: 60 + 0.9 × price. Gas: 70 + 0.4 × price.
2. Set them equal: 60 + 0.9 × price = 70 + 0.4 × price.
3. Subtract: 0.5 × price = 10, so price = 20.
4. Solar beats gas when 70 + 0.4 × price = 80, so price = 25.

Answer: gas beats coal above $20 per tonne; solar beats gas above $25 per tonne (illustrative costs)

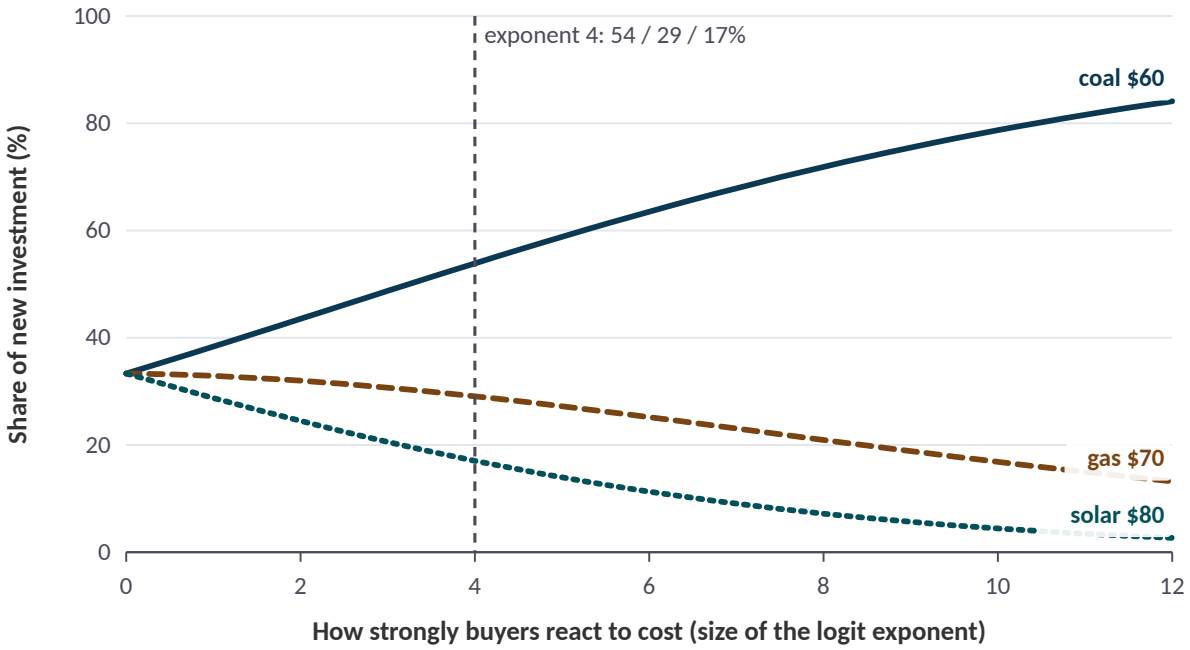


Figure 4.2 — how GCAM shares a market. What to notice: the cheapest option gets the biggest share but never all of it; how strongly buyers react to cost is itself a model setting. You will meet this rule, the logit, in Go 2.

Three: discounting. A dollar today is worth more than a dollar later, because you could invest it or use it now. Climate policy costs money now and avoids damage later, so the discount rate quietly decides many results.

$$\text{value today} = \text{future amount} \div (1 + \text{r})^{\text{years}} \tag{5}$$

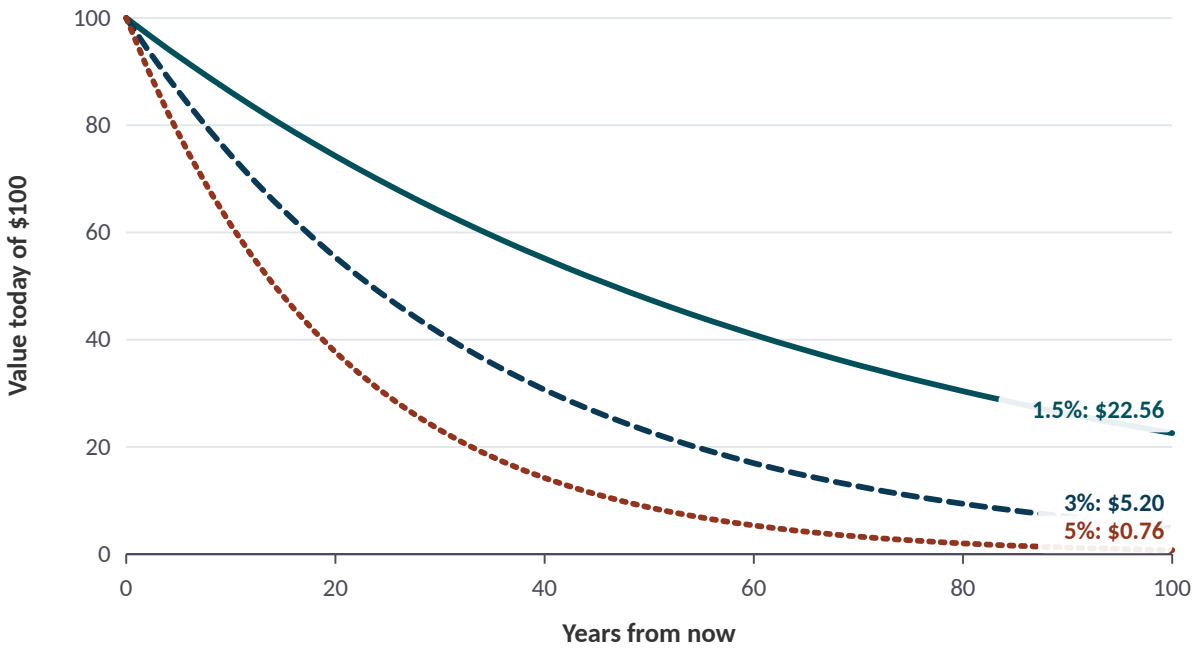


Figure 4.3 — discount curves. What to notice: the curves start together and drift apart; after a century the 5% curve has almost reached zero.

The two IAM families use these ideas differently. A cost-benefit model such as **DICE** uses discounting to weigh cutting now against damage later. A process model such as **GCAM** uses prices to decide which technologies supply each market.

TRAP "A carbon price makes everything more expensive." Only options that emit. And "a high discount rate means we care more about the future" is backwards: a high rate makes the future count **less**.

PICTURE THIS · STOP 4 OF THE ROUTE The **market square**: stallholders chalk prices on boards and rub them out until the queue in front of each stall exactly matches the pile of goods. A **toll collector** stands by the coal stall, charging coins in proportion to the smoke each sack makes. On the clock tower, coins placed at the top **shrink** as they slide down a 100-year slope.

PREDICT, THEN CHECK

Course **Toy market**: set the carbon price to **$22** per tonne. Which technology is cheapest?

My prediction	What the explorer showed

Explorer: Toy market (course site → Explore tab of week 4). **Check:** gas: coal 79.8, gas 78.8, solar 80; $22 sits between the break-even prices of $20 and $25

PREDICT, THEN CHECK

Course **Discounting explorer**: at a 5% rate, what is $100 received in 100 years worth today?

My prediction	What the explorer showed

Explorer: Discounting explorer (course site → Explore tab of week 5). **Check:** about $0.76; at 3% it is $5.20

PREDICT, THEN CHECK

In **En-ROADS** (https://en-roads.climateinteractive.org), predict what a high carbon price does to coal use by 2050, then move only the carbon-price slider.

My prediction	What the explorer showed

Explorer: En-ROADS, free. **Check:** open the graph of energy supply by source: coal should shrink most, because it has the highest emission factor (Equation 4)

TRY IT NOW Coal costs 60 and emits 0.9 t per MWh. At what carbon price does coal cost 105? (Use Equation 4.)

LOOK IT UP

- IAMC, What are IAMs? — cost-benefit vs process-based models.
- SERC Carleton teaching unit with webDICE, the social cost of carbon (the web tool itself is not confirmed to run).
- Lancaster University, SIAM 2.0, a simple IAM in a spreadsheet.

RECAP SO FAR

1. **What an IAM is:** a consistent what-if machine; inputs are assumptions; outputs are projections, not predictions.
2. **The climate chain:** warming tracks the running total of CO2, about 0.45 °C per 1,000 Gt; 500 Gt left for 1.5 °C.
3. **Energy and emissions:** CO2 = people × income per person × energy per $ × CO2 per energy; the factors multiply.
4. **Prices and choices:** prices balance markets; a carbon price adds emission factor × price; discounting shrinks the future.

CONCEPT 5 OF 6

Scenarios

Stories about society, a climate outcome, and how the IPCC sorts them

HOOK For years the highest scenario, SSP5-8.5, was called “business as usual”. It describes a world that burns far more coal than current trends suggest. A scenario's name tells you its assumptions, not the future.



A **scenario** is a consistent what-if story with numbers stapled to it, like a choose-your-own-adventure book with a spreadsheet on every page. Since about 2016 the research community has used two labels together. The **Shared Socioeconomic Pathways (SSPs)** are five stories about how society develops. They are placed by how hard they make it to cut emissions and to adapt (Figure 5.1).

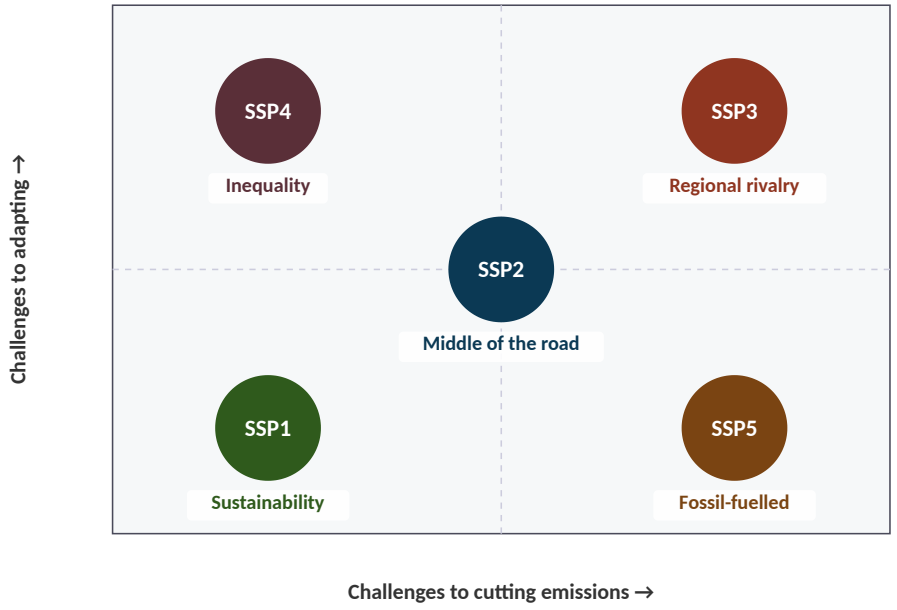


Figure 5.1 — five stories about society (after O'Neill et al., 2014). What to notice: SSP5 is easy to adapt in (rich) but hard to cut emissions in (fossil-hungry); SSP4 is the reverse.

The second label is a number: the **radiative forcing** in 2100, in W/m², inherited from the older Representative Concentration Pathways (RCPs). Higher number, more warming. A scenario name joins the two: **SSP2-4.5** is the middle-of-the-road story pushed to 4.5 W/m².

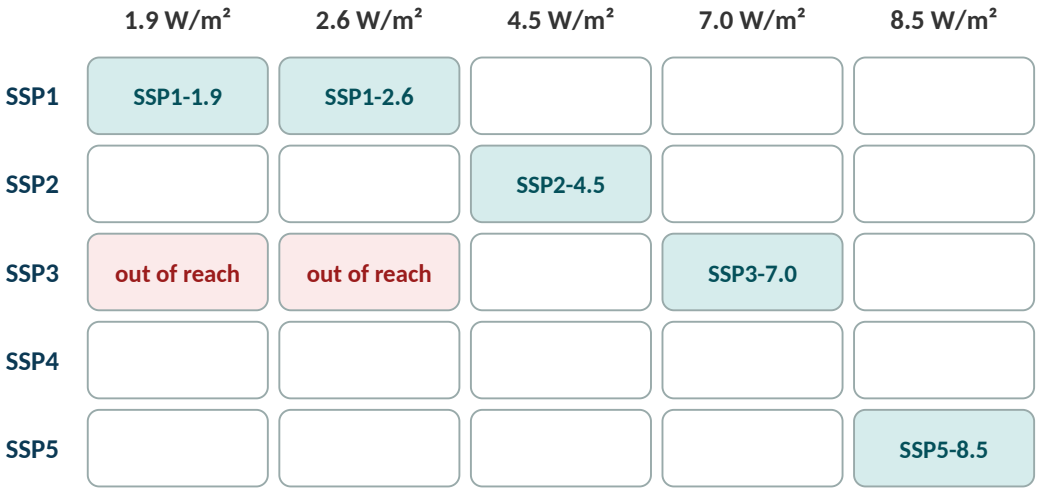


Figure 5.2 — the scenario matrix. What to notice: highlighted cells are the five headline scenarios of the IPCC's 2021 climate report; models could not reach the lowest levels in SSP3's world of regional rivalry.

SIMPLE CALCULATION

Decode SSP2-4.5

1. SSP2: the middle-of-the-road story about society.
2. 4.5: forcing in 2100 of 4.5 W/m². It is **not** 4.5 °C.
3. The IPCC's 2021 report assesses SSP2-4.5 at roughly 2.7 °C of warming by late century.
4. Its 2022 report collected about 3,000 pathways and sorted 1,202 by warming: 1,202 ÷ 3,000 ≈ 40%.

Answer: a middle-of-the-road society, 4.5 W/m² of forcing, about 2.7 °C

For its 2022 mitigation report the IPCC sorted pathways into eight **categories**, C1 to C8, by the warming they lead to (Figure 5.3).

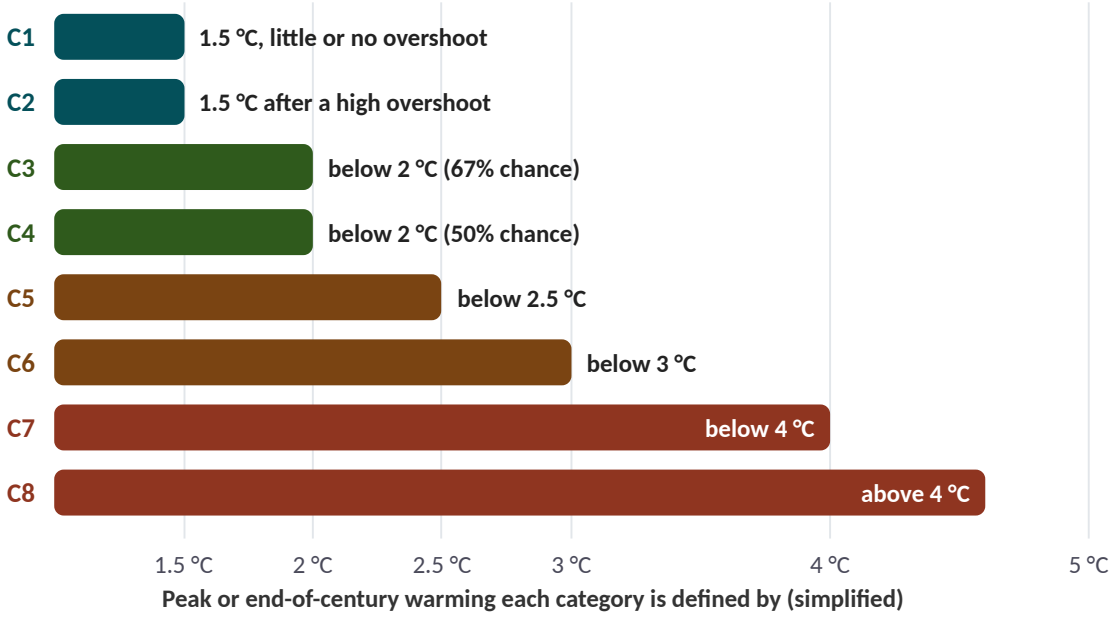


Figure 5.3 — IPCC AR6 categories, simplified. What to notice: C1 and C2 both end near 1.5 °C, but C2 overshoots first. Check the exact wording in Box SPM.1 of the report.

Scenarios are tools that get updated. In 2026 a new set was published for the next generation of climate models (CMIP7): seven scenarios, each produced by a different IAM team; the high one, H, comes from GCAM.

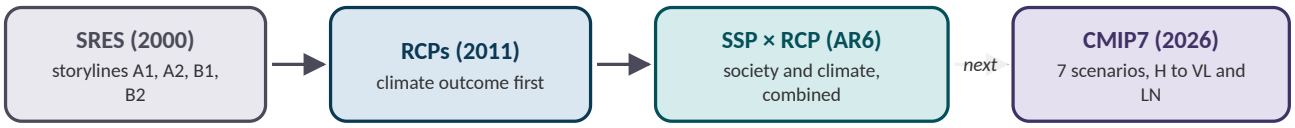


Figure 5.4 — scenario generations. What to notice: SSP × RCP and C1 to C8 are still the published IPCC framework; CMIP7 is what comes next.

TRAP "30 of 40 models show it, so it is 75% likely." No: models are not a random sample of futures, and they share data and code. A spread across models describes the tools, not the odds. Also: a **reference** scenario is a yardstick, not a prediction.

PICTURE THIS · STOP 5 OF THE ROUTE A **crossroads where five roads** meet. Each road has a signpost painted with a world: a green garden (1), a plain grey road (2), a road blocked by border fences (3), a road where gated villas look down on shacks (4), and a road paved with coal (5). On every road stands a row of heat lamps with dials marked **1.9, 2.6, 4.5, 7.0, 8.5**. At the end, a sorting office stamps travellers C1 to C8.

PREDICT, THEN CHECK

Course **SSP × RCP matrix**: click SSP3-1.9. Before reading, predict whether models found it reachable, and why.

My prediction	What the explorer showed

Explorer: SSP × RCP matrix (course site → Explore tab of week 6). **Check:** out of reach: with weak cooperation and slow technology change, models could not cut that deep

PREDICT, THEN CHECK

Open the Our World in Data IPCC Scenarios Data Explorer. Predict which story has the most people in 2100, and whether SSP1 and SSP5 differ in population.

My prediction	What the explorer showed

Explorer: Our World in Data IPCC scenarios explorer, free, no log-in. **Check:** SSP3 is highest by far; SSP1 and SSP5 share a low-population path that peaks and then falls, even though their energy systems are opposites

PREDICT, THEN CHECK

Optional, instructor demo: the IIASA AR6 Scenario Explorer. Predict when CO_2 reaches net zero in C1 pathways compared with C3, then plot Emissions|CO2 for both.

My prediction	What the explorer showed

Explorer: AR6 Scenario Explorer, IIASA (shows a log-in link; test access before class). **Check:** C1 pathways reach net-zero CO2 earlier, around mid-century; C3 pathways later

TRY IT NOW Decode SSP5-8.5 and SSP1-2.6 in one sentence each. Which would you call a reference, and why is that word risky?

LOOK IT UP

- Carbon Brief, Explainer: How Shared Socioeconomic Pathways explore future climate change.
- Carbon Brief, What the new IPCC report says about how to limit warming to 1.5C or 2C — the C1 to C8 table.
- Carbon Brief, Explainer: the CMIP7 emissions scenarios (1 September 2026).
- IPCC AR6 WGIII, Summary for Policymakers, Box SPM.1.

RECAP SO FAR

1. **What an IAM is:** a consistent what-if machine; inputs are assumptions; outputs are projections, not predictions.
2. **The climate chain:** warming tracks the running total of CO2, about 0.45 °C per 1,000 Gt; 500 Gt left for 1.5 °C.
3. **Energy and emissions:** CO2 = people × income per person × energy per $ × CO2 per energy; the factors multiply.
4. **Prices and choices:** prices balance markets; a carbon price adds emission factor × price; discounting shrinks the future.
5. **Scenarios:** SSP = story about society, the RCP number = forcing in 2100; AR6 sorts pathways C1 to C8.

CONCEPT 6 OF 6

One model, end to end

A tour of GCAM, and how to read what it says

HOOK The highest of the seven new CMIP7 scenarios, published in 2026, was produced with GCAM, the Global Change Analysis Model from the Pacific Northwest National Laboratory. This concept is a tour of that model, with no code.



GCAM splits the world into **32 energy-economy regions**, more than 300 land units and **235 river basins**. It starts from a calibrated base year, **2021**, and then steps forward **five years at a time to 2100**. In each step an auctioneer adjusts the prices of fuels, crops, land and water until supply equals demand in every market. Then the clock moves on. GCAM never peeks ahead: decisions use today's prices only (it is **recursive-dynamic**). Its emissions go to **Hector**, a simple climate model, which returns concentrations, forcing and temperature.

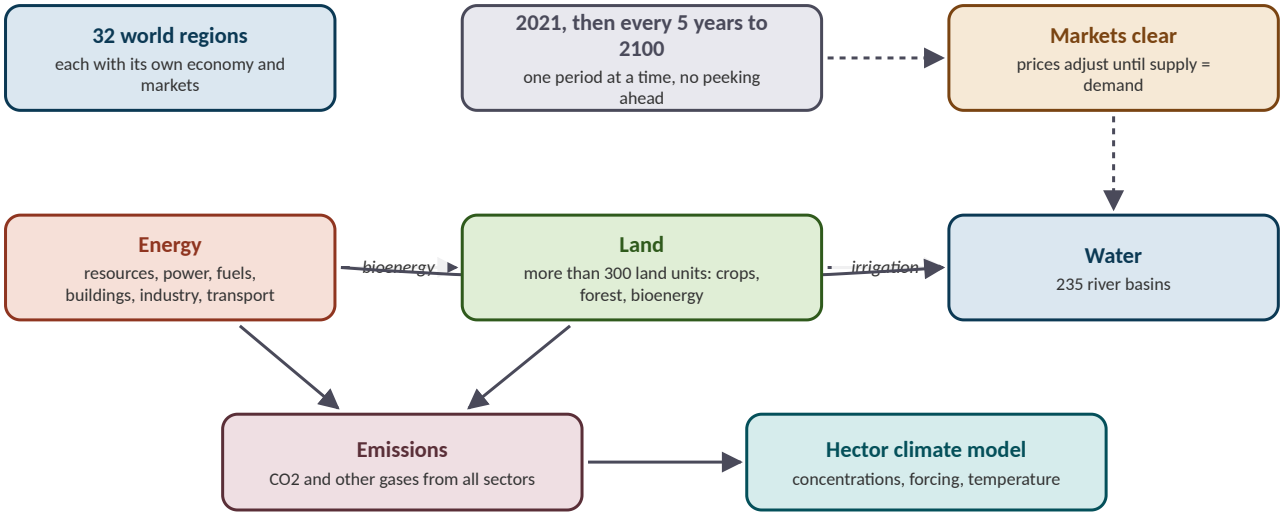


Figure 6.1 — a tour of GCAM. What to notice: energy, land and water are linked by bioenergy, irrigation and power-plant cooling; all of them feed emissions; emissions feed Hector. Source: GCAM v8.2 documentation; the latest release is GCAM 9.1 (June 2026).

How do you read a GCAM result? Never alone. Every study runs a **Reference** scenario (no new climate policy) and one or more **policy** scenarios with exactly the same inputs plus one change. The effect of the policy is the **difference**.

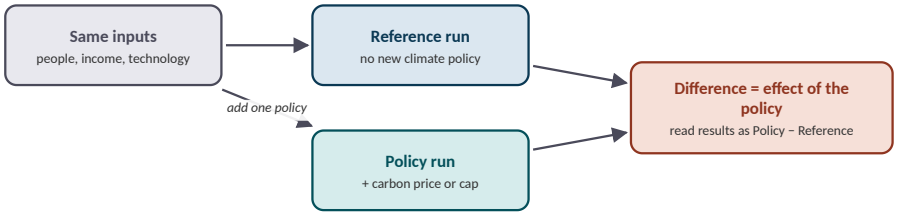


Figure 6.2 — reference vs policy. What to notice: the reference is a yardstick, not a forecast; the finding is the gap between the two runs.

A policy can enter the model in two ways. A **carbon price** is a toll road: you fix the cost per tonne and the model finds how much emissions fall. An **emissions limit** is a gate: you fix the emissions and the model finds the price that achieves it. On the same cost curve they are mirror images.

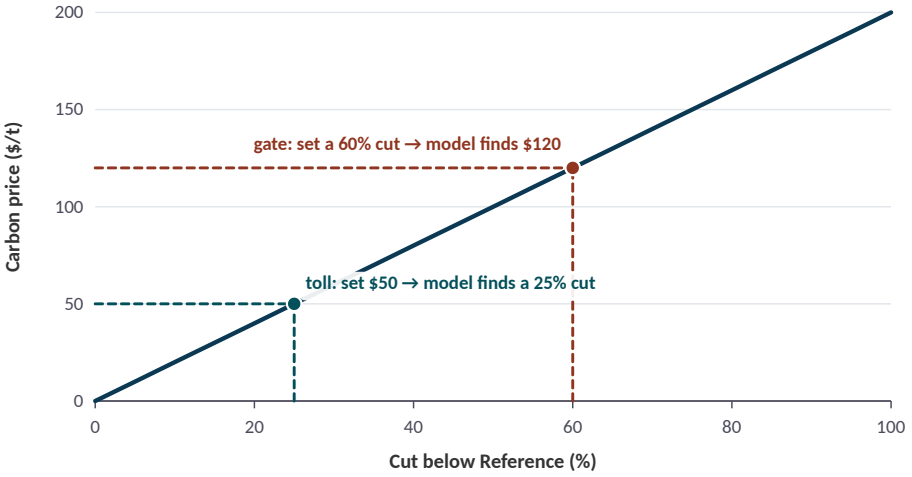


Figure 6.3 — price and cap are mirrors (a straight-line cost curve, price = 2 × cut, for illustration). What to notice: you can enter at either axis and read off the other.

SIMPLE CALCULATION

Toll or gate, and what if carbon removal falls short?

1. Toll: set $50 per tonne; the model finds the cut: 50 ÷ 2 = 25% below Reference.
2. Gate: require a 60% cut; the model finds the price: 2 × 60 = $120 per tonne.
3. A pathway plans to remove 200 Gt of CO_2 but only half is delivered: 100 Gt more stays in the air.
4. Concept 2's rule: 0.45 °C × 100 ÷ 1,000 = 0.045 °C of extra warming.

Answer: $50 → 25% cut; 60% cut → $120; undelivered removal of 100 Gt adds about 0.045 °C

What IAMs cannot tell you matters as much. Four criticisms every user should know: heavy **reliance on carbon removal** later in the century; the cheapest cuts are found but **who pays** is left open; models are **hard to inspect** (open code like GCAM's helps); and **assumptions age**, for example past solar cost estimates that proved too high. One review's verdict: keep the models, question them, and supplement them.

TRAP "GCAM has a button for each country's climate pledge." It does not: turning a pledge into a price or a limit is the modeller's choice, and a good report says how it was done. And "the models disagree, so they are useless" is wrong: the reasons they disagree (structure, inputs, method) are findings in themselves.

PICTURE THIS · STOP 6 OF THE ROUTE The walk ends at a **control tower** with **32 screens**. An auctioneer shouts prices until every screen turns green; a stopwatch beeps every **5 years** and he moves on. From the window you see two runways: on one, every plane pays a **toll**; on the other, a **gate** lets exactly 60 through. A twin tower opposite (Reference) shows the same screens without the toll: the story is in the difference.

PREDICT, THEN CHECK

Course **Carbon price vs cap explorer**: start the price at $25 in 2025. What yearly growth rate do you need for a 60% cut by 2050?

My prediction	What the explorer showed

Explorer: Carbon price explorer (course site → Explore tab of week 10). **Check:** about 6.5% a year, because the price must reach $120 in 2050: (120 ÷ 25)^(1/25) − 1

PREDICT, THEN CHECK

Open the IIASA ScenarioMIP (CMIP7) explorer. Predict whether the H scenario (GCAM) or the VL scenario (REMIND-MAgPIE) has more coal in 2050, then look.

My prediction	What the explorer showed

Explorer: ScenarioMIP Explorer, IIASA, pre-release (shows a log-in link; test access first). **Check:** H, the high scenario, keeps far more fossil fuel; VL is very low emissions. Different teams, different stories, same reporting template

TRY IT NOW Your friend says: “GCAM showed that a $50 carbon price cuts emissions by 25%.” Write the two questions you would ask before believing it.

LOOK IT UP

- GCAM v8.2 documentation, GCAM model overview — read the first two screens and Figure 1.
- PNNL, GCAM – Global Change Analysis Model.
- IAMC wiki, GCAM reference card (describes GCAM 7.0).
- Gambhir et al. (2019), A review of criticisms of integrated assessment models.

RECAP SO FAR

1. **What an IAM is:** a consistent what-if machine; inputs are assumptions; outputs are projections, not predictions.
2. **The climate chain:** warming tracks the running total of CO2, about 0.45 °C per 1,000 Gt; 500 Gt left for 1.5 °C.
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4. **Prices and choices:** prices balance markets; a carbon price adds emission factor × price; discounting shrinks the future.
5. **Scenarios:** SSP = story about society, the RCP number = forcing in 2100; AR6 sorts pathways C1 to C8.
6. **One model, end to end:** GCAM clears markets for 32 regions every 5 years; read Policy − Reference; price and cap are mirrors.

Master mind map

All six concepts on one page. Cover the leaves with your hand and say them aloud; then redraw the map from memory.



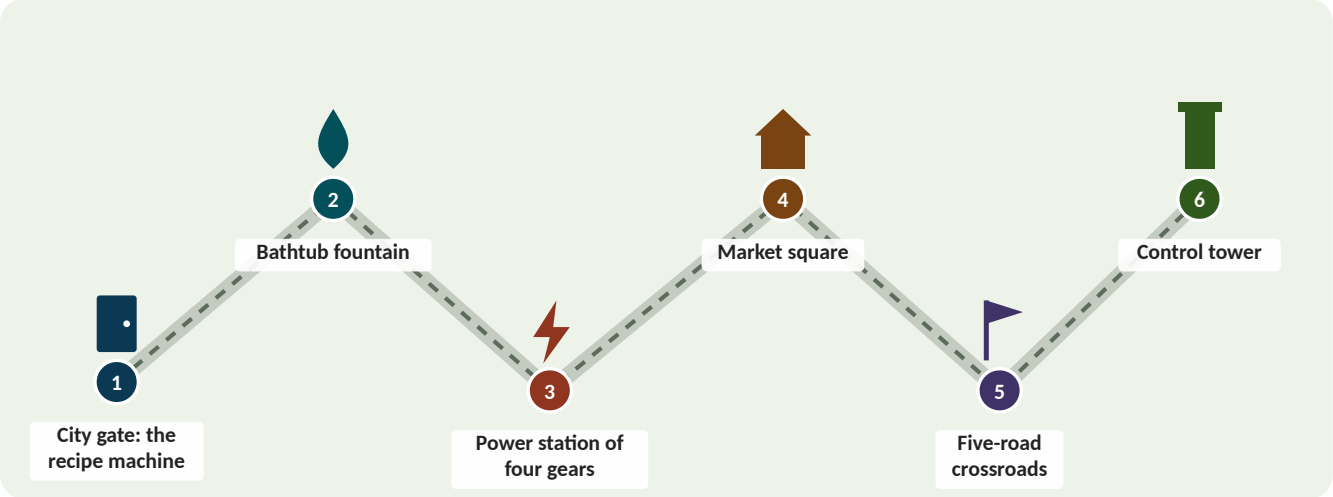
Connections and contrasts

Pair students confuse	The one fact that separates them
Projection vs prediction	A projection says what follows *if* the assumptions hold; a prediction claims what *will* happen. IAMs make projections.
Cost-benefit vs process-based IAM	Cost-benefit asks how much to cut (DICE); process-based asks how a goal could be met (GCAM).
Emissions vs concentration	Emissions are a yearly flow (Gt per year, the tap); concentration is the stock in the air (ppm, the water level).
Net zero vs zero	Net zero: remaining emissions equal removals. Zero: nothing emitted.
Adding vs multiplying changes	Kaya factors multiply: Avalon is ×1.008, not +20%.
Carbon price vs emissions cap	Price fixed, emissions found (toll); cap fixed, price found (gate).
SSP vs RCP number	SSP is the story about society; the number is forcing in 2100 in W/m², not °C.
High vs low discount rate	A higher rate makes the future count less.
Reference vs policy scenario	Reference is a yardstick; the policy's effect is the difference between the two runs.
Model agreement vs probability	A spread across models describes the tools, not the odds.

Symbols and units

Symbol	Meaning	Where
Gt CO_2	billion tonnes of carbon dioxide	1, 2, 6
ppm	parts per million of CO_2 in the air	2
W/m²	watts per square metre: extra heat trapped (forcing)	2, 5
°C	warming above pre-industrial	2, 5, 6
EJ	exajoule, 10^{18} joules of energy	3
MWh	megawatt-hour of electricity	4
$/t	dollars per tonne of CO_2 (GCAM itself reports per tonne of carbon)	4, 6
r	discount rate per year	4

The route: a walk through a model city



The route. Six stops, one per concept, in learning order.

1. City gate: the recipe machine
Inputs (people, coins, a solar panel, a law book) go in; a consistent cake with a thermometer candle comes out. Two gatekeepers: scales (cost-benefit) and toolbox (process-based).

2. Bathtub fountain
Tap 40 buckets a year, corked drain, red rim at 500. Each 1,000 buckets lifts the thermometer 0.45 degrees.

3. Power station of four gears
People, coins, a leaky pipe, a chimney brush, geared together so they multiply. Sign: 100 in, 36 home.

4. Market square
Chalk prices rubbed out until queues match piles. A toll collector charges by the smoke. Coins shrink sliding down the 100-year clock.

5. Five-road crossroads
Garden, grey road, fences, villas over shacks, coal-paved road. Heat lamps dialled 1.9 to 8.5. A sorting office stamps C1 to C8.

6. Control tower
32 screens turn green; a stopwatch beeps every 5 years. Toll runway vs gate runway; a twin Reference tower opposite.

WALK-THROUGH DRILL Close the guide. Walk the six stops in your head. At each one say the idea and one number: consistency (50% cut); 0.45 °C per 1,000 Gt and 500 Gt; 100 in, 36 home and ×1.008; $20 and $25 break-evens and $5.20; 4.5 W/m² is not 4.5 °C; 32 regions, 5 years, $50 → 25%.

Quiz

Twenty questions in three tiers: warm-up 6, core 10, challenge 4. Pass mark for Go 1: 14 of 20.

Warm-up

1. An integrated assessment model is best described as:
   A. a forecast of the year 2100
   B. a consistent what-if calculator for energy, economy, land and climate
   C. a weather model run for longer
   D. a tool that chooses climate policy

2. Which of these is an **input** to an IAM?
   A. warming in 2100
   B. coal used in 2050
   C. a carbon price
   D. the price of electricity

3. The climate responds mainly to:
   A. this year's CO_2 emissions
   B. the total CO_2 emitted so far
   C. the price of oil
   D. the number of models

4. In SSP2-4.5, the 4.5 means:
   A. 4.5 °C of warming
   B. 4.5 W/m² of forcing in 2100
   C. 4.5 billion people
   D. 4.5 Gt of CO_2

5. A carbon price changes the cost of solar power by:
   A. nothing, its emission factor is zero
   B. the full carbon price
   C. half the carbon price
   D. it depends on the discount rate

6. Label boxes A to G of the blank machine below from memory.

Core

7. The world emits 40 Gt a year and 500 Gt remain for 1.5 °C. If emissions fall in a straight line to zero, the budget lasts:
   A. 12.5 years
   B. 20 years
   C. 25 years
   D. 50 years

8. Using 0.45 °C per 1,000 Gt, emitting 2,000 Gt more adds about:
   A. 0.45 °C
   B. 0.9 °C
   C. 2 °C
   D. 4.5 °C

9. People ×1.1, income per person ×2, energy per $ ×0.5, CO_2 per energy ×0.9. CO_2 changes by a factor of:
   A. 0.99
   B. 1.1
   C. 1.5
   D. 4.5

10. Avalon's factors change by +20%, +50%, −30% and −20%. Its CO_2:
    A. rises 20%
    B. rises about 0.8%
    C. falls 20%
    D. is unchanged exactly

11. Coal 60 $/MWh and 0.9 t/MWh; gas 70 $/MWh and 0.4 t/MWh. Gas becomes cheaper than coal above a carbon price of:
    A. $10
    B. $20
    C. $25
    D. $50

12. At a 3% discount rate, $100 in 100 years is worth today about:
    A. $97
    B. $52
    C. $5.20
    D. $0.76

13. Fill the gaps in the climate chain below (boxes 2 and 3).

14. 12 of 18 models show fast solar growth. You can conclude:
    A. it is 67% likely
    B. most models behave that way under these assumptions
    C. the other 6 are wrong
    D. it is certain

15. In GCAM, a carbon price of $50 is an example of:
    A. an output the model finds
    B. an input you set, from which the model finds emissions
    C. a climate target
    D. a share weight

16. How should a GCAM policy result be read?
    A. as a forecast
    B. as the difference between the policy run and the Reference run
    C. as the Reference run alone
    D. as an average of all runs

Challenge

17. Find the three errors in the broken diagram below and correct them.

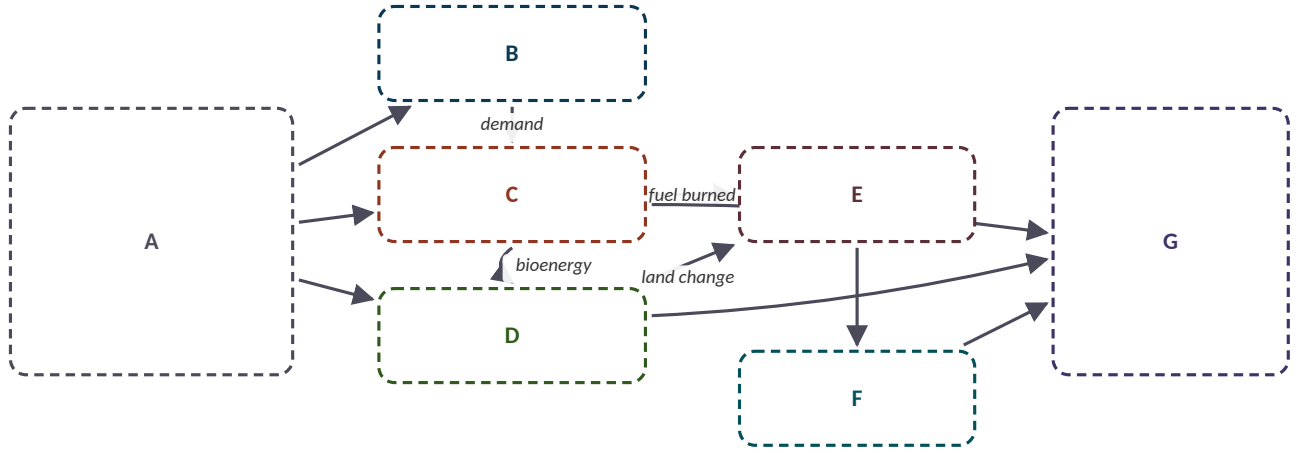
18. Explain in two sentences why a gradual fall in emissions buys twice the time of a sudden stop for the same budget.

19. A pathway plans 300 Gt of CO_2 removal but only a third is delivered. Extra warming:

    A. 0.045 °C
    B. 0.09 °C
    C. 0.135 °C
    D. 0.3 °C

20. Explain why SSP5 sits in the corner “hard to cut, easy to adapt”, and name one GCAM policy input that would push an SSP5 world towards C3.

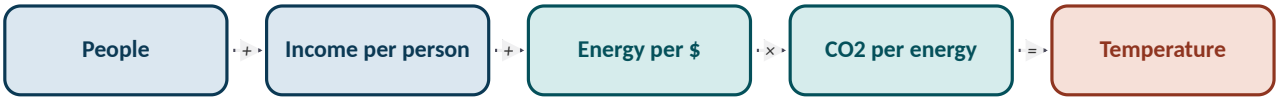
Exercise visuals



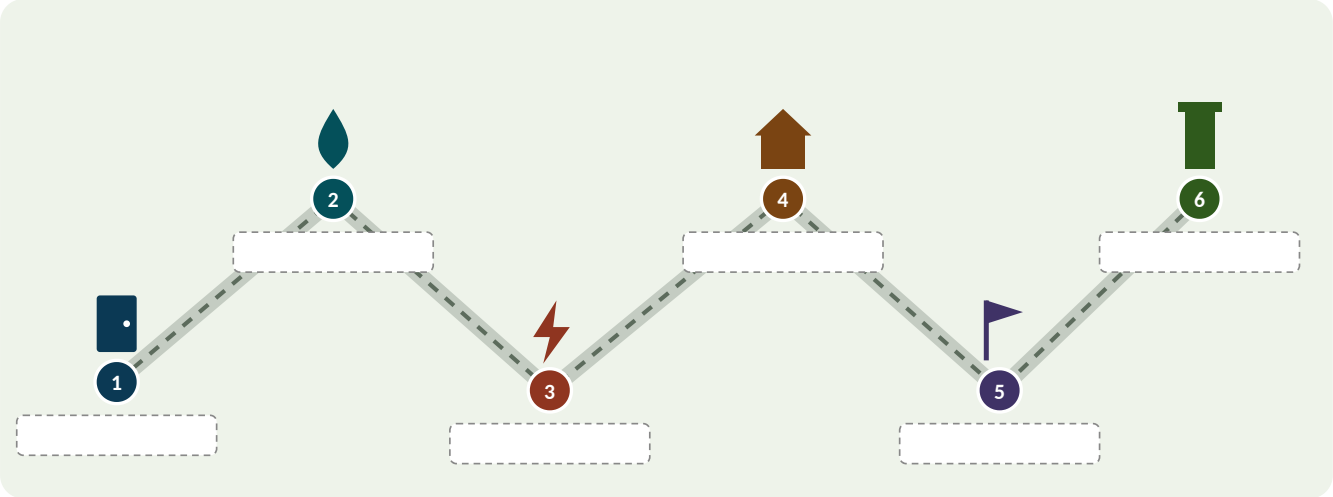
Question 6 — label boxes A to G.



Question 13 — fill boxes 2 and 3.



Question 17 — this Kaya chain has three errors.



Bonus — name the six stops of the route.

Score routing

If you missed	Reread
1, 2, 6, 15, 16	Concept 1 and Concept 6
3, 7, 8, 13, 18, 19	Concept 2
9, 10, 17	Concept 3
5, 11, 12	Concept 4
4, 14, 20	Concept 5

Make it stick

BUILD THE SMALLEST WORKING EXAMPLE (15 MINUTES)

On paper, build a one-line IAM for your own country. Pick a population, an income per person, an energy per dollar and a CO_2 per energy (from Our World in Data), and compute today's CO_2 with the Kaya identity. Then invent a 2050 story by changing each factor, compute 2050 CO_2, add up 30 years of emissions with a straight line between the two years, and turn the total into warming with 0.45 °C per 1,000 Gt.

Self-check with Avalon's numbers: 600 Mt CO_2 today and ×1.008 by year 30 give 604.8 Mt. A straight line between them sums to about 30 × (600 + 604.8) ÷ 2 = 18,072 Mt, i.e. 18.1 Gt, adding about 0.008 °C.

TEACH IT TO A 14-YEAR-OLD (10 MINUTES) Explain what an IAM is and why the IPCC uses them, without these words: model, scenario, emissions, equilibrium, forcing. Record yourself; if you used a banned word, start again.

FILL THE BLANK ROUTE Use the blank route on the quiz page. For each stop write the concept, the scene and one number.

Revision plan

When	What	Done
Today	Read one concept; do its two Predict, then check boxes	☐
Day 1	Walk the route; redo the Try it now boxes from memory	☐
Day 3	Cheat sheet with the right-hand half covered	☐
Day 7	Quiz, warm-up and core tiers; follow the routing table	☐
Day 14	Whole quiz; build the smallest working example	☐
Before Go 2	Master mind map from memory; one En-ROADS scenario that reaches 2 °C	☐

Cheat sheet

1. What an IAM is

Consistent what-if machine. Inputs: people, income, technology, resources, policies. Outputs: energy, land, prices, emissions, temperature. Two families: cost-benefit (DICE), process-based (GCAM). *Stop 1: recipe machine at the gate.*

2. The climate chain

Emissions → ppm → W/m² → °C. ΔT ≈ 0.45 °C per 1,000 Gt (likely 0.27–0.63). Budget from 2020: 500 Gt (1.5 °C, 50%), 1,150 Gt (2 °C, 67%). 40 Gt/yr: 12.5 yrs flat, 25 yrs straight line. *Stop 2: bathtub fountain.*

3. Energy and emissions

Primary → conversion → final → services. 100 coal → 40 out of plant → 36 at home. CO2 = P × GDP/P × E/GDP × CO2/E. Factors multiply: Avalon 1.2×1.5×0.7×0.8 = 1.008. *Stop 3: four gears.*

4. Prices and choices

Equilibrium: supply = demand (100 → 127 $/MWh with tax). New cost = base + ef × price: coal 105, gas 90, solar 80 at $50. Break-evens $20 and $25. PV = F ÷ (1+r)^t: $100 in 100 yrs = $22.56 / $5.20 / $0.76 at 1.5 / 3 / 5%. *Stop 4: market square.*

5. Scenarios

SSP1 sustainability, SSP2 middle, SSP3 rivalry, SSP4 inequality, SSP5 fossil-fuelled. Number = W/m² in 2100. SSP2-4.5 ≈ 2.7 °C. AR6: about 3,000 pathways, 1,202 sorted C1–C8. CMIP7 (2026): 7 scenarios, H by GCAM. *Stop 5: five-road crossroads.*

6. One model: GCAM

32 regions, 2021 base, 5-year steps to 2100, more than 300 land units, 235 basins, recursive-dynamic, markets clear, Hector climate. Read Policy – Reference. Toll $50 → 25% cut; gate 60% → $120. Limits: removal reliance, equity, transparency, ageing assumptions. *Stop 6: control tower.*

Numbers to remember: 40 Gt a year · 500 Gt · 0.45 °C per 1,000 Gt · 36 of 100 · ×1.008 · $20 and $25 · $5.20 · 4.5 W/m² ≈ 2.7 °C · 32 regions, 5 years.

Answer key

Try it now

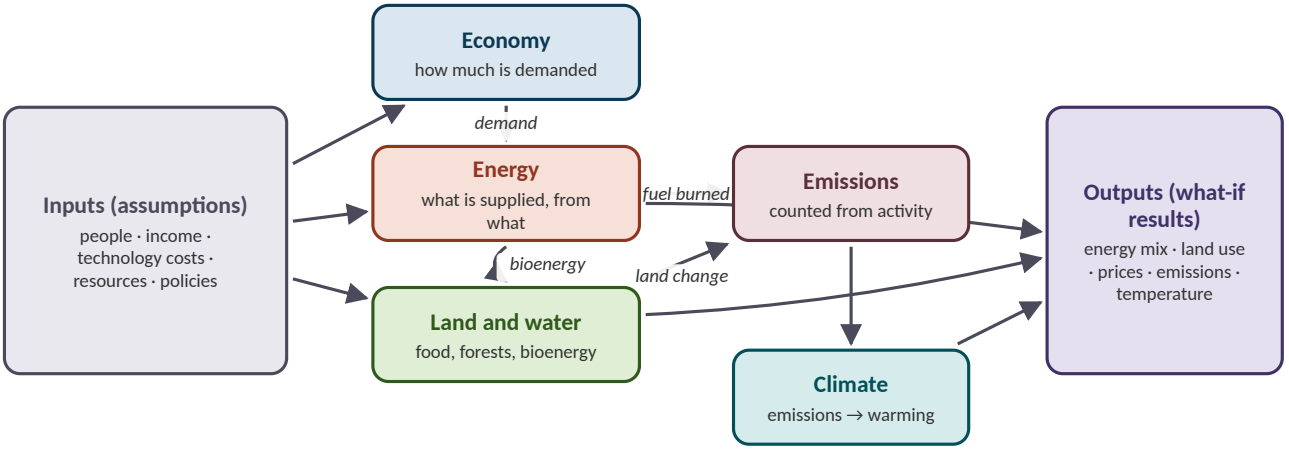
- **Concept 1:** inputs (a), (c), (e); outputs (b), (d), (f).
- **Concept 2:** 0.45 × 800 ÷ 1,000 = 0.36 °C; 800 ÷ 40 = 20 years of today's emissions.
- **Concept 3:** 2 × 2 × 0.5 × 0.5 = 1: CO_2 is unchanged.
- **Concept 4:** 60 + 0.9 × price = 105, so 0.9 × price = 45 and price = $50.
- **Concept 5:** SSP5-8.5: fossil-fuelled growth reaching 8.5 W/m², a very high-emission world. SSP1-2.6: a sustainable world with strong climate policy reaching 2.6 W/m². Neither is a forecast; “reference” suggests “what will happen”, which no scenario is.
- **Concept 6:** (1) Compared with which Reference, and what else differed between the runs? (2) Which assumptions (costs, demand, removal) drive the 25%, and was a sensitivity test done?

Predict, then check: computed values

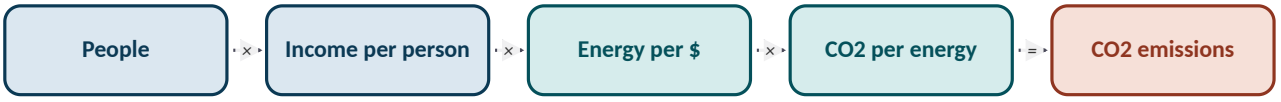
- Warming explorer: zero in 2050 gives 751 Gt and 1.44 °C; zero in 2070 gives 1,171 Gt and 1.63 °C.
- Bathtub at 30 Gt: 16.7 and 33.3 years. Kaya with energy per $ ×0.5: 0.72. Toy market at $22: gas. Discounting at 5%: $0.76. Carbon price growth for 60%: 6.5% a year.

Quiz

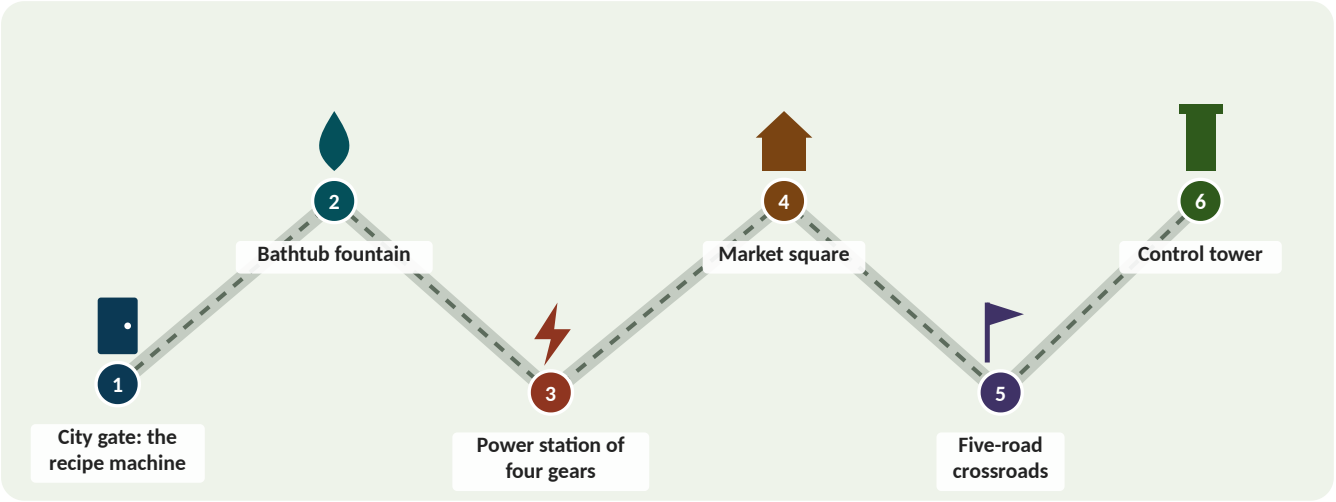
1. B (§1)
2. C (§1): the others are outputs
3. B (§2)
4. B (§5)
5. A (§4)
6. A inputs, B economy, C energy, D land and water, E emissions, F climate, G outputs (§1, completed figure on the Connections page)
7. C: ½ × 40 × T = 500 gives T = 25 (§2). A is the flat case
8. B: 0.45 × 2 = 0.9 °C (§2)
9. A: 1.1 × 2 × 0.5 × 0.9 = 0.99 (§3)
10. B: 1.2 × 1.5 × 0.7 × 0.8 = 1.008 (§3). A comes from adding percentages
11. B: 0.5 × price = 10 (§4)
12. C: 100 ÷ 1.03^100 = 5.20 (§4); D is the 5% answer
13. Box 2 concentration (ppm), box 3 forcing (W/m²) (§2)
14. B (§5): models are not a random sample
15. B (§6)
16. B (§6)
17. Errors: the first two signs should be × not +, and the last box should be CO2 emissions, not temperature; temperature comes later, through the climate chain (§3, corrected figure below)
18. Equal budgets mean equal areas under the emission curves. A straight line from 40 to zero has half the height on average, so it lasts twice as long (§2)
19. B: two thirds undelivered = 200 Gt; 0.45 × 200 ÷ 1,000 = 0.09 °C (§6)
20. SSP5 is rich and technologically capable (easy to adapt) but runs on abundant fossil fuel (hard to cut). A carbon price rising over time, or an emissions limit, entered as a policy input (§5, §6)



Question 6, completed machine.



Question 17, corrected.



Bonus, completed route.

Going further

Core readings

1. Westphal, M. I. (2021). *Greenhouse Gas Emissions Scenarios: Background, Issues, and Policy Relevance* (CRS Report R46807). Congressional Research Service. https://www.everycrsreport.com/reports/R46807.html — the most beginner-friendly.
2. Weyant, J. (2017). Some contributions of integrated assessment models of global climate change. *Review of Environmental Economics and Policy*, 11(1), 115–137. https://doi.org/10.1093/reep/rew018
3. O'Neill, B. C., Kriegler, E., Riahi, K., et al. (2014). A new scenario framework for climate change research: the concept of shared socioeconomic pathways. *Climatic Change*, 122, 387–400. https://doi.org/10.1007/s10584-013-0905-2
4. Riahi, K., van Vuuren, D. P., Kriegler, E., et al. (2017). The Shared Socioeconomic Pathways and their energy, land use, and greenhouse gas emissions implications: an overview. *Global Environmental Change*, 42, 153–168. https://doi.org/10.1016/j.gloenvcha.2016.05.009
5. Keppo, I., Butnar, I., Bauer, N., et al. (2021). Exploring the possibility space: taking stock of the diverse capabilities and gaps in integrated assessment models. *Environmental Research Letters*, 16(5), 053006. https://doi.org/10.1088/1748-9326/abe5d8
6. van Vuuren, D. P., O'Neill, B. C., Tebaldi, C., et al. (2026). The Scenario Model Intercomparison Project for CMIP7 (ScenarioMIP-CMIP7). *Geoscientific Model Development*, 19, 2627–2656. https://doi.org/10.5194/gmd-19-2627-2026 (abstract and scenario table).

Simulators and explorers

- En-ROADS https://en-roads.climateinteractive.org and its tutorial; C-ROADS and the World Climate role-play https://www.climateinteractive.org/world-climate-simulation-resources.
- The Global Calculator https://www.globalcalculator.net — lifestyle and demand levers, to 2050.
- Our World in Data IPCC scenarios explorer https://ourworldindata.org/explorers/ipcc-scenarios; SSP Scenario Explorer https://ssp.apps.ece.iiasa.ac.at/.
- Climate Action Tracker thermometer https://climateactiontracker.org/global/cat-thermometer/ — where current policies are heading.
- Videos: AR6 Scenario Explorer by Edward Byers https://www.youtube.com/watch?v=HFgkS5oQbH4; UNED introduction to IAMs (English subtitles) https://canal.uned.es/video/5c9e2151a3eeb03f118b4568.

HONESTY NOTE

Illustrative, invented for teaching: the recipe-calculator energy numbers, Avalon, the 40% plant and 10% grid losses, the coal, gas and solar costs, the supply and demand curves, the straight-line cost curve (price = 2 × cut) and the 200 Gt removal example. **From published sources:** the 0.45 °C per 1,000 Gt estimate and its likely range, the 500 and 1,150 Gt budgets from 2020 (IPCC AR6 WGI, 2021); about 3,000 pathways and 1,202 categorised C1–C8 (IPCC AR6 WGIII, 2022, via Carbon Brief); SSP2-4.5 at about 2.7 °C (AR6 WGI); the SSP challenge space (O'Neill et al., 2014); CMIP7 scenarios and H produced with GCAM (Carbon Brief, 1 September 2026; ScenarioMIP explorer); GCAM structure (GCAM v8.2 documentation; latest release GCAM 9.1, June 2026). The course explorers are toy models written for this course, not GCAM. Simulator pages were checked on 23 October 2026; several IIASA explorers show a log-in link, so test them before class.